



# Supply Chain & Financial Analytics



## Project Overview

This project presents a **Power BI dashboard** that delivers a unified view of business performance across financial and operational domains. It enables stakeholders to monitor key metrics, identify trends, and make **data-driven decisions** efficiently.



## Business Objective

- Monitor revenue and profitability trends
- Evaluate supply chain and supplier performance
- Identify inventory shortages and risks
- Track shipment efficiency and delivery performance
- Analyze customer behavior and contribution



## Data Source

The dashboard is built using multiple structured CSV datasets:

- Sales
- Customer
- Product
- Supplier
- Inventory
- Procurement
- Production
- Shipment
- Facility
- Date Table



## Data Preparation

- Performed data cleaning and transformation using **Power Query**
- Designed a **Star Schema data model** for efficient analysis
- Created a centralized **Date Table** for time-based reporting
- Handled missing values and ensured data consistency



## Key KPIs

- Total Revenue
- Gross Revenue
- Profit
- Profit Margin %
- Total Orders
- Inventory Quantity
- Shipment Quantity
- On-Time Delivery %
- Total Customers



## Dashboard Pages

### 1 Index Page

Serves as a navigation hub to access all dashboard sections.

### 2 Executive Overview

Provides a high-level summary of business performance using key KPIs and revenue trends.

### 3 Supply Analysis

Analyzes supplier performance, procurement costs, and contribution to overall operations.

### 4 Inventory Analysis

Tracks stock levels across products and locations, highlighting low inventory risks.

### 5 Shipment Analysis

Evaluates shipment trends, delivery performance, and identifies delay patterns.

### 6 Customer Analysis

Explores customer segmentation, top customers, and purchasing behavior.



## Filters & Interactivity

- Date-based filtering
- Product category selection
- Customer and supplier filters
- Cross-filtering enabled across all visuals for dynamic analysis

## DAX Measures

Total Revenue = SUM(Sales[net\_revenue])

Gross Revenue = SUM(Sales[gross\_revenue])

Profit = SUM(Sales[profit])

Profit Margin % = DIVIDE([Profit], [Total Revenue], 0)

Total Orders = DISTINCTCOUNT(Sales[order\_number])

Avg Order Value = DIVIDE([Total Revenue], [Total Orders], 0)

## Key Insights

- Revenue trends provide visibility into overall business growth
- Supplier analysis highlights high-cost and high-performing vendors
- Inventory analysis helps identify potential stock shortages
- Shipment analysis reveals delivery efficiency and delay issues
- Customer analysis identifies key contributors to revenue

## Conclusion

This dashboard provides a **comprehensive and interactive view of business performance**, enabling stakeholders to make informed decisions across financial, operational, and customer domains.

## Screenshots



